How to compute sign charts quickly

2023 Fall

The method

Given a rational function (polynomial over polynomial),

1. Put it into factored form.
2. Draw a number line and put tick marks for each zero and vertical asymptote (places where numerator=0 and where denominator = 0
3. Going from right to left, determine the signature of each subinterval.

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Why this is fast:

Crossing a tick mark will only affect one factor, flipping its sign. Whether that affects the sign of the entire function depends on what power the factor was raised to.

$$(-)^{\text{even}} = + \qquad \text{so} \qquad (+)^{\text{even}} \rightsquigarrow (-)^{\text{even}} \quad \text{doesn't flip}$$

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Why start from the right

Starting at the right will make all factors positive, unless you've written things oddly. One usually writes $-(x-3)$, rather than $(3-x)$. The sign of the rightmost interval is simply determined by the leading coefficients.

Worked example

Example

Determine the sign chart for the following function:

$$\frac{-4(x-3)^7(x-2)^8(x+3)^4}{x^6(x-4)^5(x+1)^3}$$

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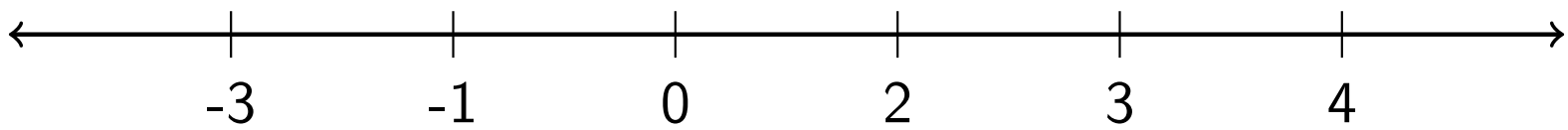

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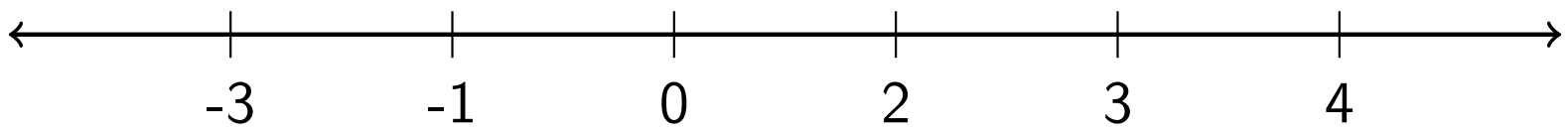

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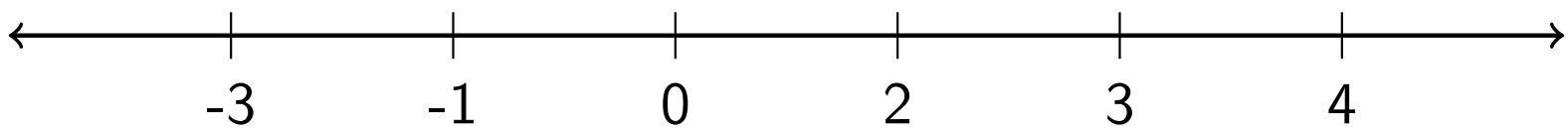

Starting at the right, each factor is positive. The "-4" in the front makes the sign '$-$':

$$\frac{-4(+)^7(+)^8(+)^4}{(+)^6(+)^5(+)^3}$$

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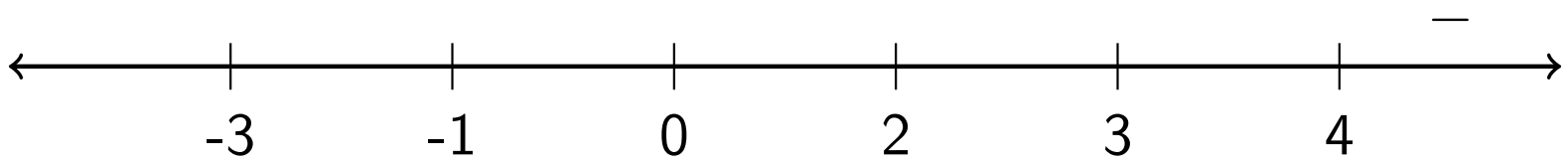

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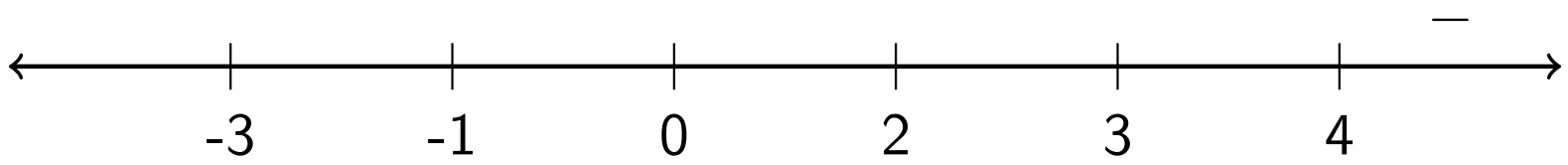

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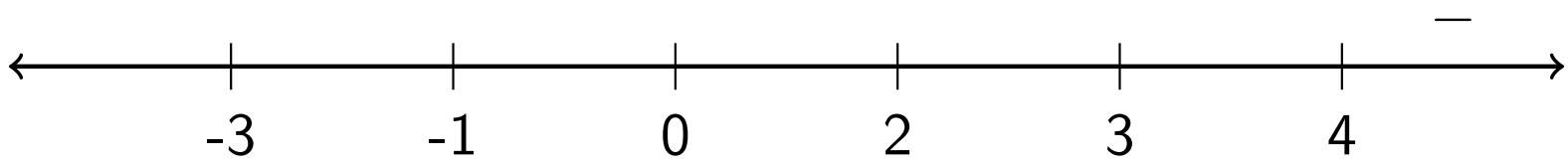

Moving to the left, we first cross $x = 4$, which corresponds to the factor $(x - 4)$. This flips the factor. It's power is 5, so the sign *does* flip on the number line.

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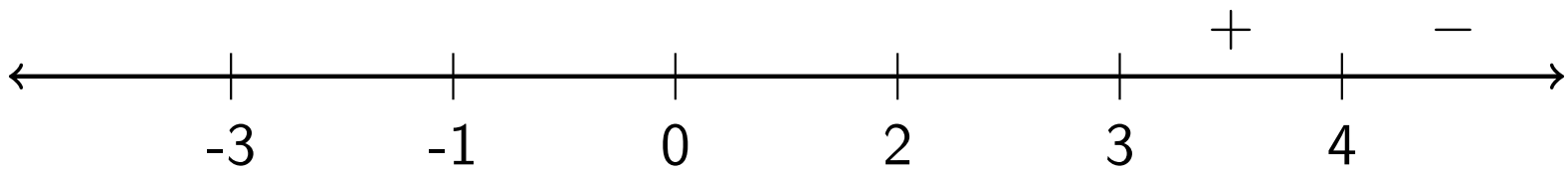

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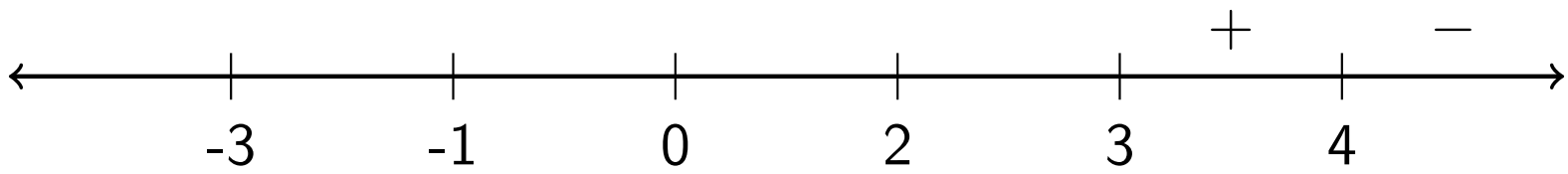

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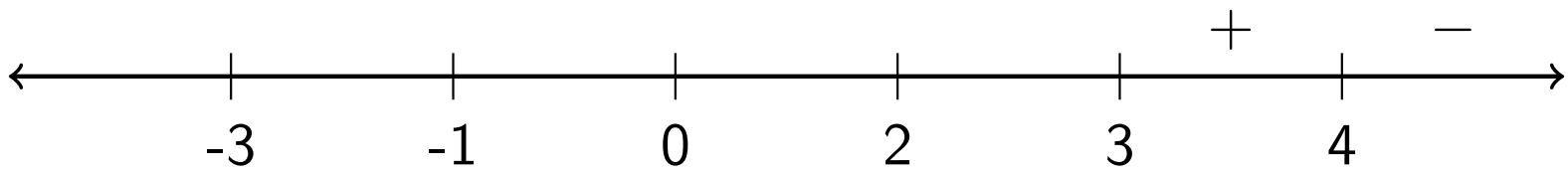

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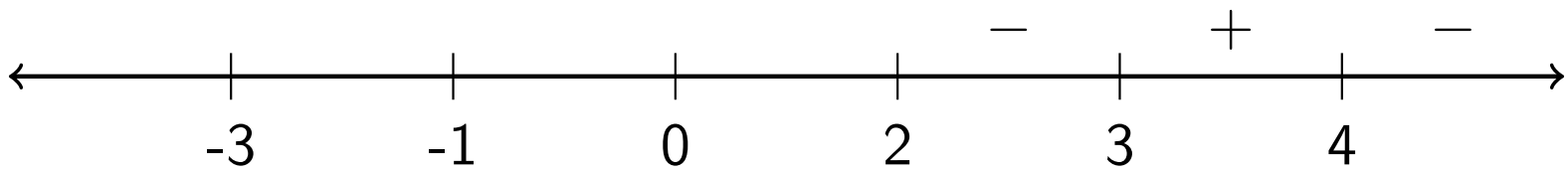

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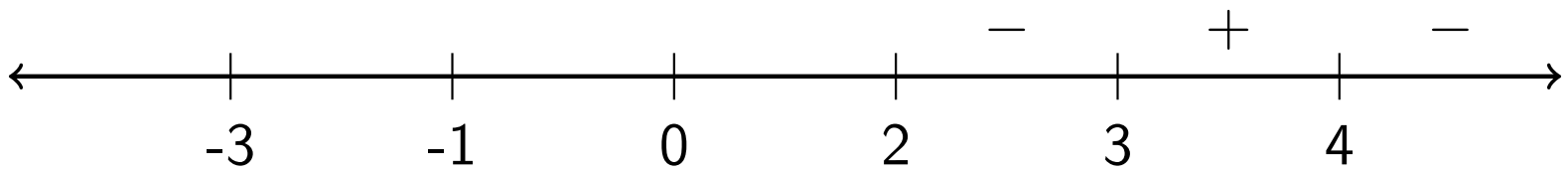

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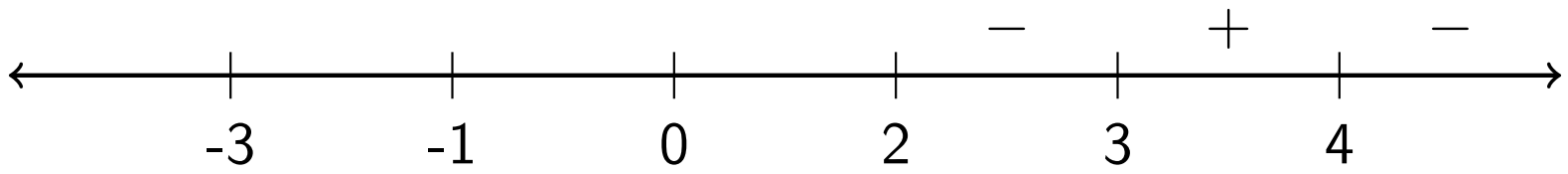

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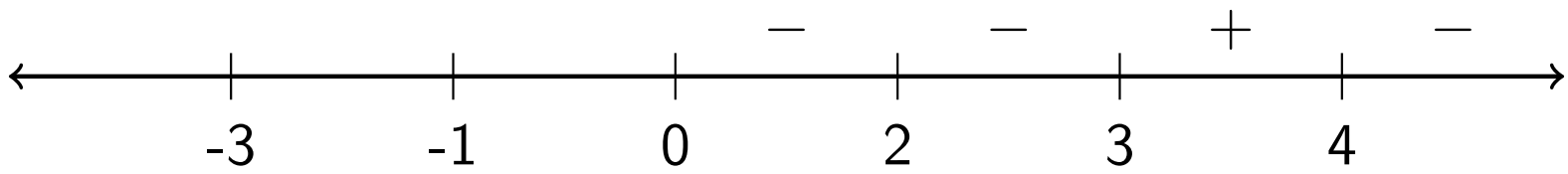

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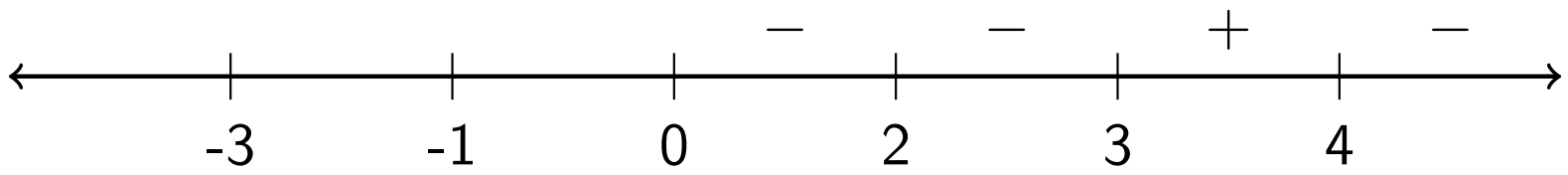

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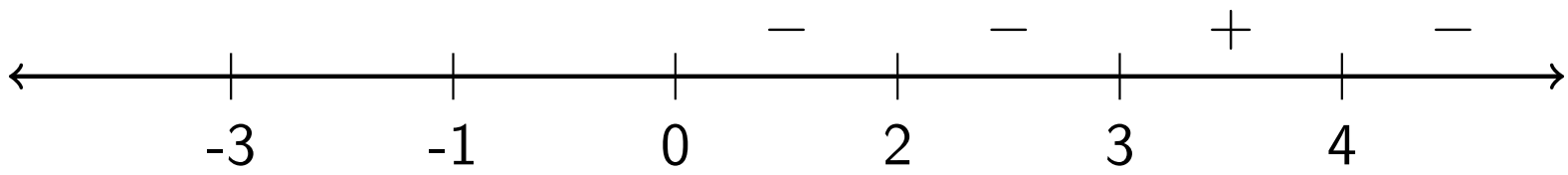

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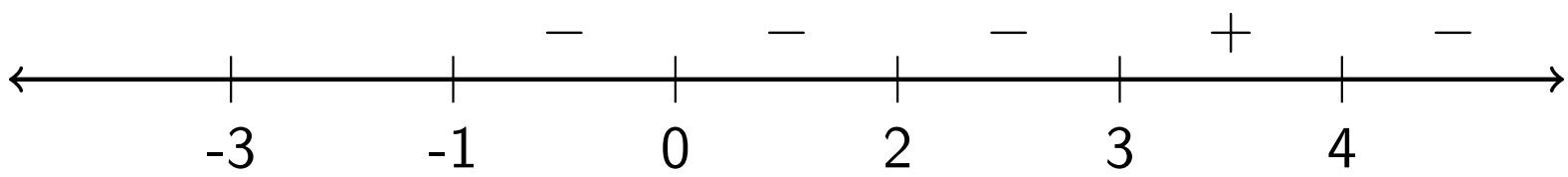

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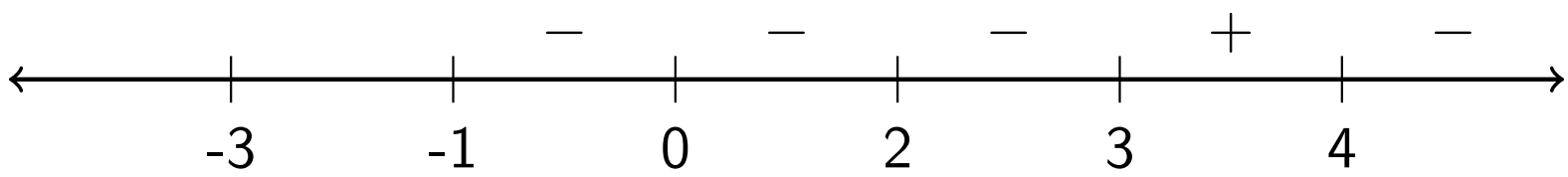

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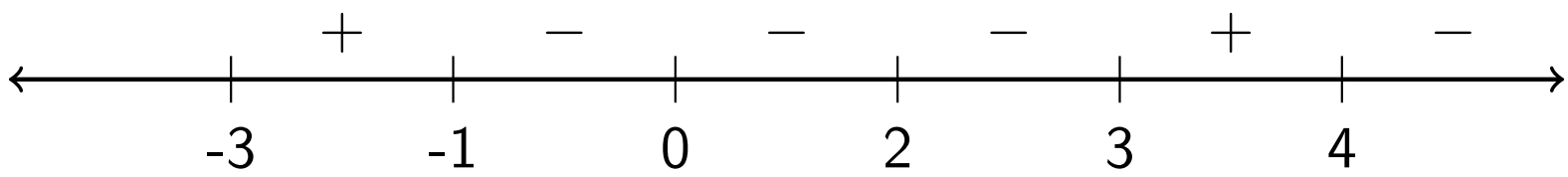

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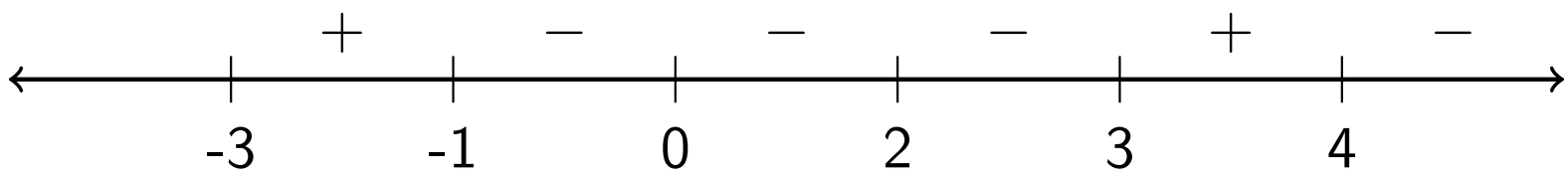

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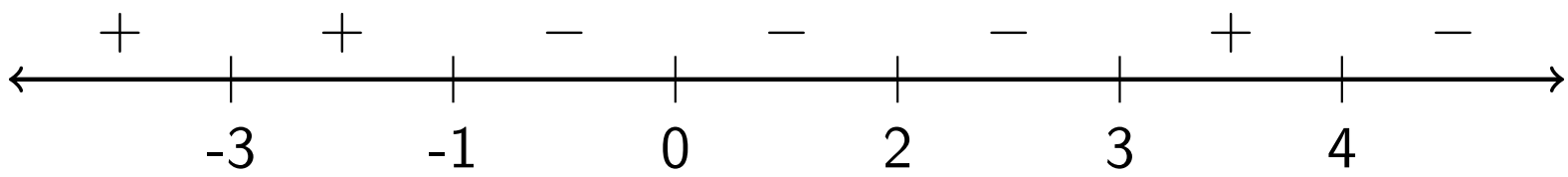

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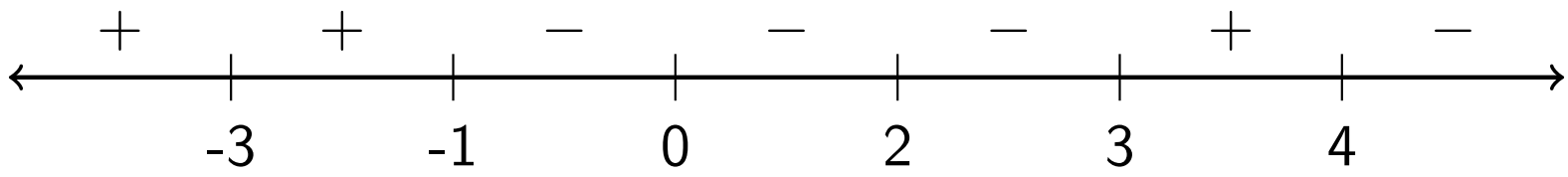
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